

Roller Conveyor – Parametric CAD Generation Rules

For Fusion 360 API Add-in (strict spec for automated generation)

These rules exist to eliminate two recurring defects: **non-constant side rail thickness** and **non-constant leg thickness**. Any generated model that violates a rule below is invalid and must be regenerated, not visually patched.

1. Core Principle

Thickness and cross-section are fixed constants. Only length/position parameters may vary with user input.

Every part has exactly two kinds of dimensions:

- **Fixed (constant) dimensions** — wall thickness, extrusion cross-section, pin diameter, rail profile. These are set once from a lookup/standard table and never derived from conveyor length, width, or roller count.

- **Variable dimensions** — conveyor length, conveyor width, number of rollers, leg height. These may change per user input, but must never feed back into a fixed dimension above.

If a generation step ever computes a thickness or cross-section as a function of length/width/count, that is a bug.

2. Side Rails

- Side rail is an extruded profile (channel/L-section) swept along the **length** axis only.
- **Wall thickness is a single named parameter** (e.g. `railThickness = 3 mm`) applied uniformly to the whole cross-section sketch. It must be defined **once**, in one sketch, and referenced everywhere else — never re-entered or recalculated per instance.
- The cross-section sketch must be extruded with a **constant profile along the entire rail length** (single extrude operation, no tapering, no per-segment scaling).
- **Inner face rule:** the inner face of the side rail must be coincident with (touching, zero gap) the outer face/end of the roller pins/shafts that protrude from each roller. This is a mate/joint constraint, not a visual guess:

- $\text{RailInnerFaceOffset} = \text{RollerBodyLength} / 2$ from the conveyor centerline, on both sides.
- Roller pin length is only what protrudes *beyond* this inner face into the rail's pin hole — it must not determine rail spacing; rail spacing determines how far the pin needs to reach.
- Left and right rails must be mirrored copies of the **same** profile/parameter set — never independently dimensioned.

3. Legs

- Legs are standard **aluminum extrusion profiles** (e.g. 20×20, 40×40 T-slot), selected from a fixed catalog of allowed cross-sections — never a custom/derived rectangle.
- **Cross-section size is a single fixed parameter for the whole model**, independent of conveyor width, length, roller count, or leg height.
- Only the **extrude length (height)** may vary, and only along the leg's own axis.
- All legs in one assembly must reference the **same** cross-section parameter instance. No leg may be generated from a separately-scaled or separately-sketched profile.

- Leg-to-rail attachment point (bracket/face) must be defined as a fixed offset from the rail's fixed profile — not recomputed from rail thickness each time.

4. Rollers

- Roller body diameter and pin diameter are fixed catalog parameters, not derived from rail spacing.
- Roller body length = $\text{ConveyorWidth} - 2 \times \text{RailInnerWallOffset} - 2 \times \text{PinEngagementClearance}$.
- Pin length is fixed (catalog value); only the **quantity** of rollers changes with conveyor length (via fixed pitch spacing), never their geometry.
- Roller pitch (center-to-center spacing) is a fixed parameter; roller count = $\text{floor}((\text{ConveyorLength} - \text{endClearance}) / \text{pitch}) + 1$. Never solve pitch backward from a desired count unless the user explicitly overrides pitch.

5. Assembly / Joint Rules

- Use Fusion **joints** or **as-built joints**, not manual positioning by coordinate math, for:
 - Roller pin → rail pin-hole (rigid/pin joint, zero gap).
 - Leg top face → rail bottom face (rigid joint).

- Every joint must reference the **named fixed parameters** above (rail thickness, leg cross-section, pin diameter) — never a one-off numeric literal typed into a joint offset.
- No component may be scaled (non-uniform scale, "Scale" feature) to force a fit. If a part doesn't fit, the parameter values are wrong — fix the parameter, not the geometry after the fact.

6. Parameter Naming Convention (required)

All fixed dimensions must exist as **User Parameters** in Fusion, named and typed exactly as:

Parameter	Type	Notes
<code>railThickness</code>	fixed	single source of truth for both rails
<code>railProfileHeight</code>	fixed	rail cross-section height

Parameter	Type	Notes
<code>legProfile</code>	fixed (enum/catalog)	e.g. "40x40"
<code>rollerBodyDia</code>	fixed	
<code>rollerPinDia</code>	fixed	
<code>rollerPinLength</code>	fixed	
<code>rollerPitch</code>	fixed (default) or user- override	
<code>conveyorLength</code>	variable	user input
<code>conveyorWidth</code>	variable	user input
<code>legHeight</code>	variable	user input

Variable parameters may only appear in **sketches that define length/position**. Fixed parameters may only appear in **cross-section sketches**. A generation script must never write to a fixed parameter inside a loop that iterates over length/count.

7. Pre-Generation Validation

Checklist (run before every output)

1. Is `railThickness` referenced from exactly one parameter, on both rails? /
2. Is `legProfile` identical across every leg instance? /
3. Does the rail inner face coincide with roller pin outer face (zero gap, joint-verified)? /
4. Are any parts using non-uniform scale? (must be none) /
5. Were any fixed dimensions computed inside the length/count loop? (must be none) /
6. Do left/right rails and all legs derive from the same profile instance, not duplicated/re-sketched copies? /

If any check fails, do not emit the model — fix the parameter reference and regenerate.